

Seagrass as Bioremediation Candidate for Filtering Microplastics

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1. Introduction and Background

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- 1.2. How is it released into the environment?
- 1.3. Why is this an issue?

2. Current conventional treatment methods

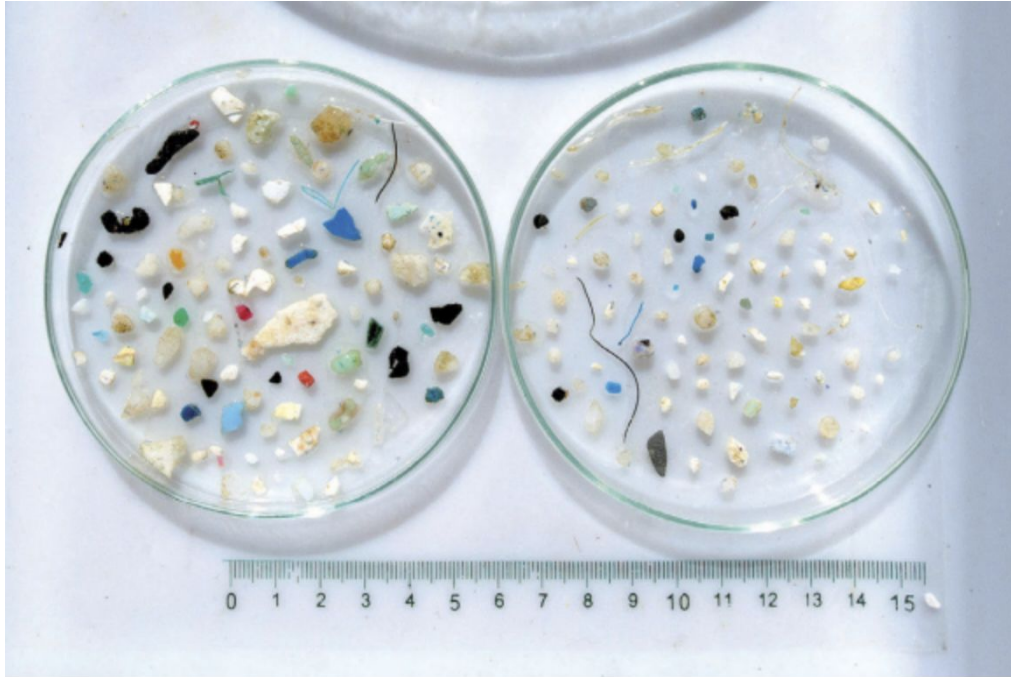
- 2.1. Improving existing facility
- 2.2. Membrane technology and biofiltration
- 2.3. Bioremediation and its challenges

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1. Introduction and Background

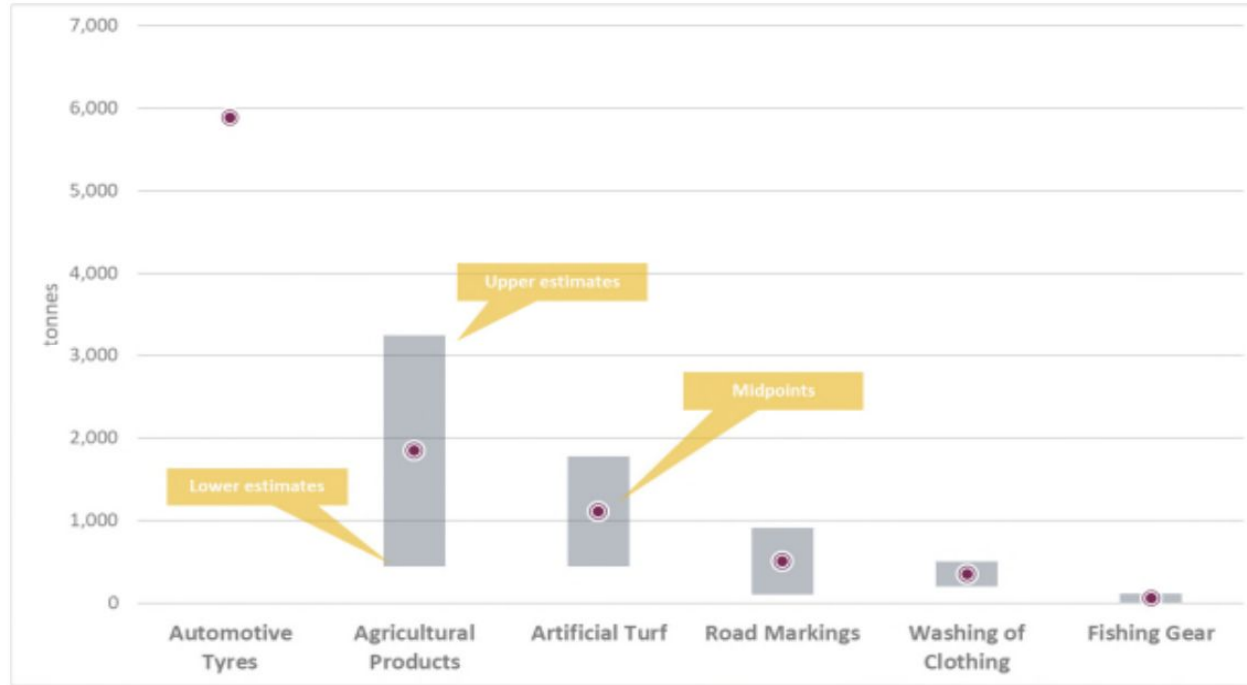
1.1. What is Microplastic?



- Plastic particles of sizes <5mm to 0.1µm.
- Primary - e.g. microbeads
Secondary - e.g. degraded plastic bottle
- Transported mainly via water
- WWTP main source

Main Sources of MPs in Scotland

Figure 1: Summary of Microplastic Source Emission Estimates for Scotland for Priority Products



- Automotive Tyres
- Agricultural Products
- Artificial Turf
- Road Markings
- Washing Clothes

1.2. How is it released into the environment?

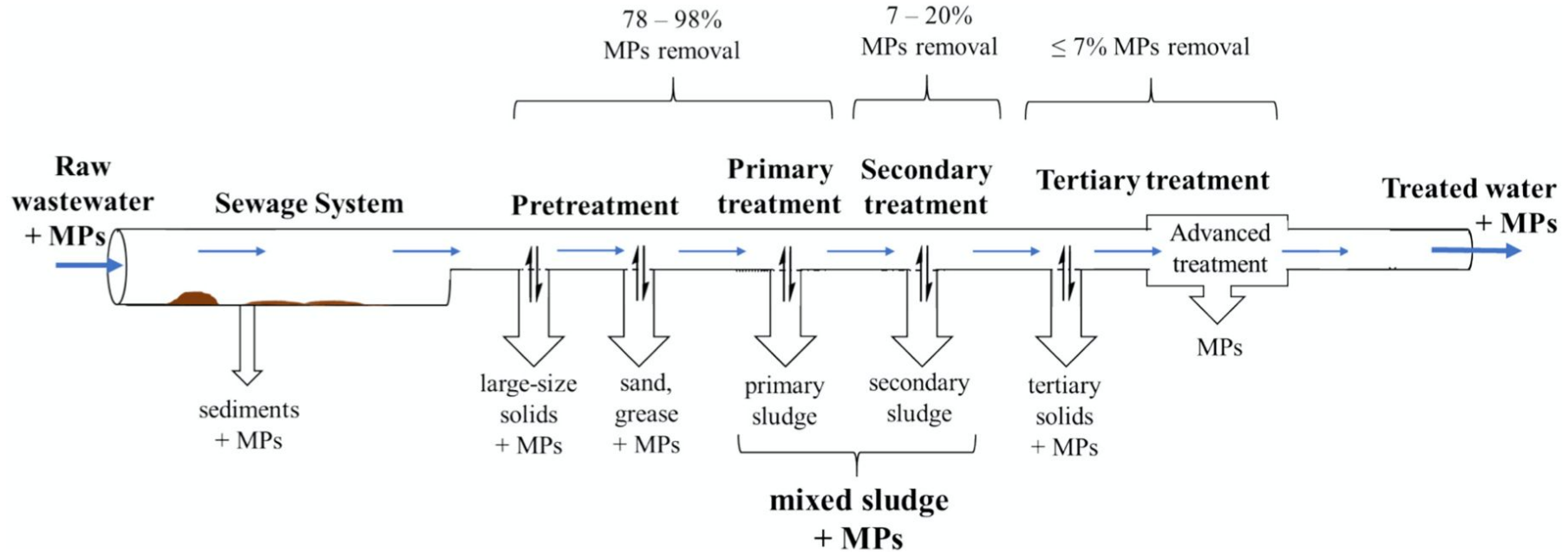
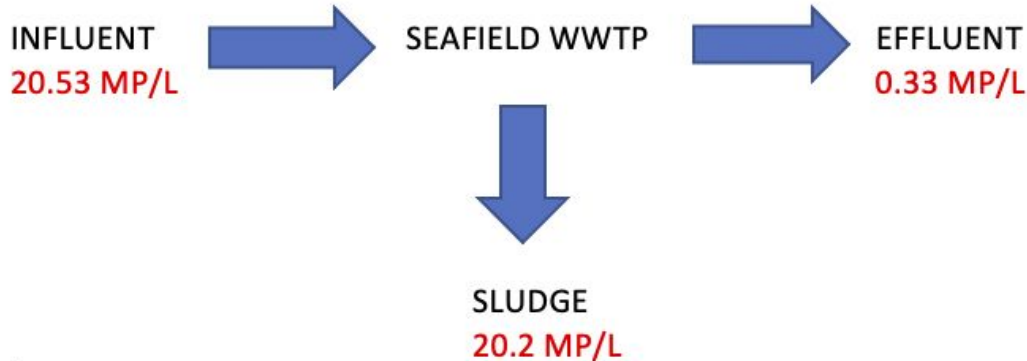


Fig. 1. A schematic representation of WWTP processes and percentages of MPs removal during processing.

Estimation of MPs in Edinburgh

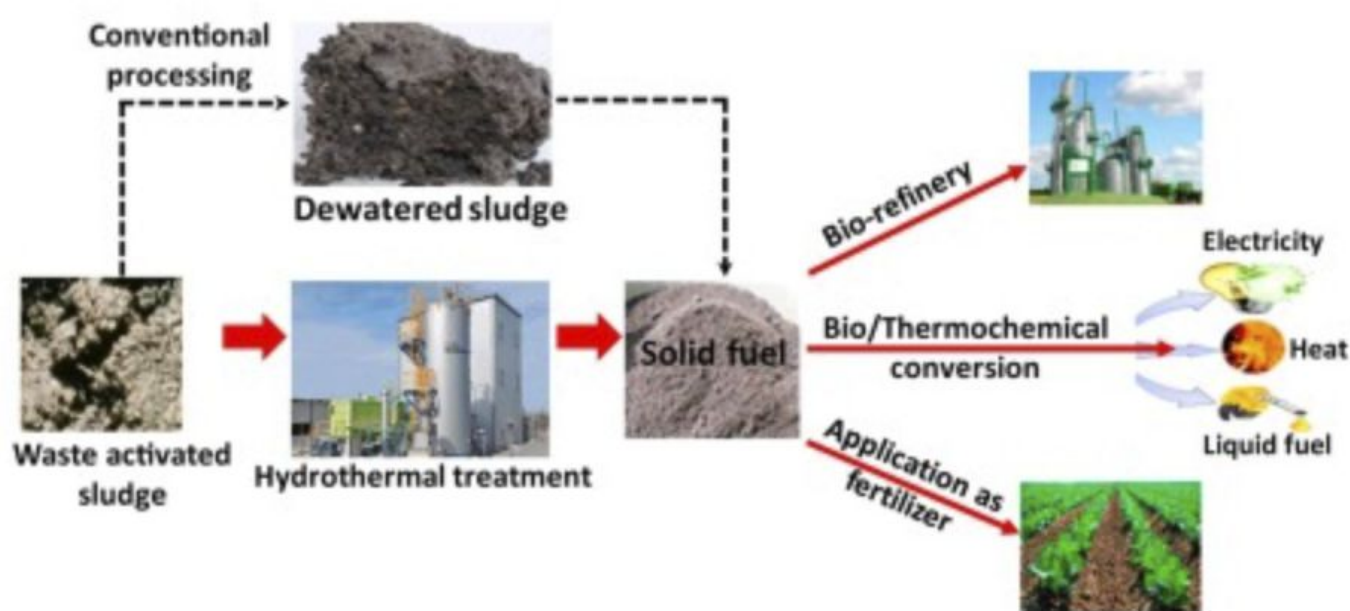
- Seafeld WWTP annually produces ~30,000 tonnes of biosolid fertiliser for agriculture.
- Data extrapolated from a study in Glasgow measuring MP concentration:



- Amount of MPs generated per day from Seafeld = **6.06×10^9 MPs**

Murphy *et al.*, 2016

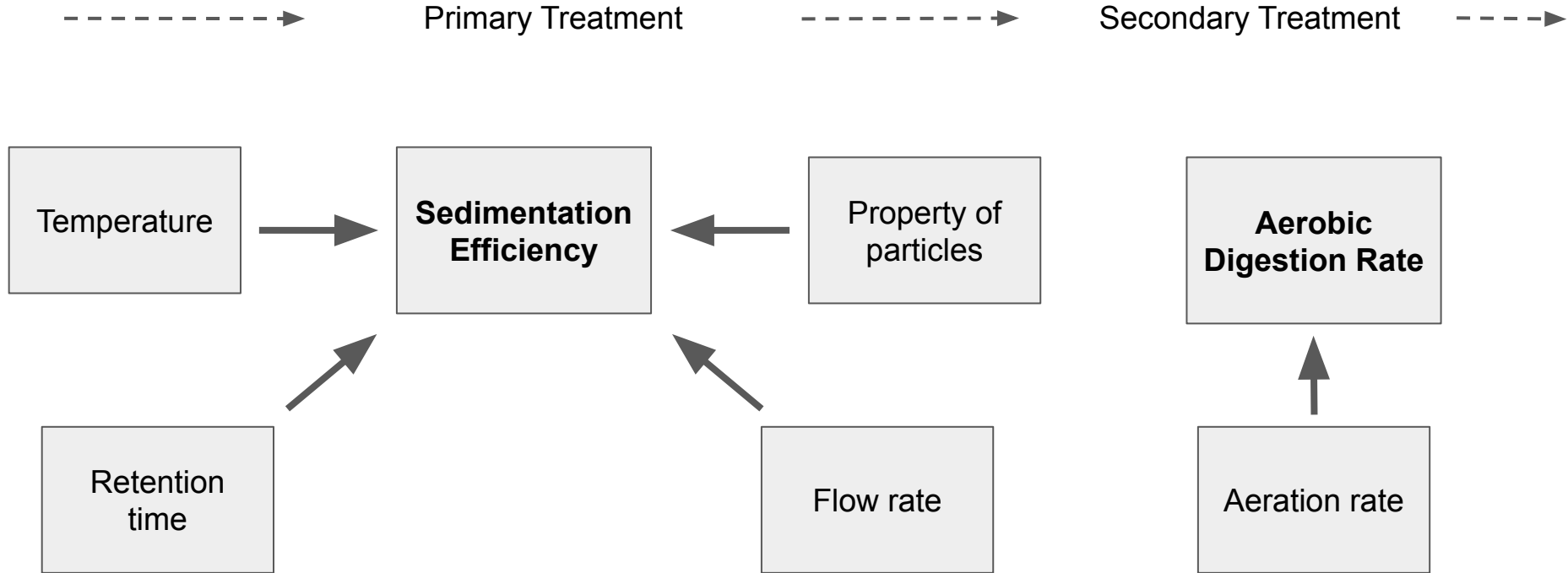
1.3. Why is this an issue?



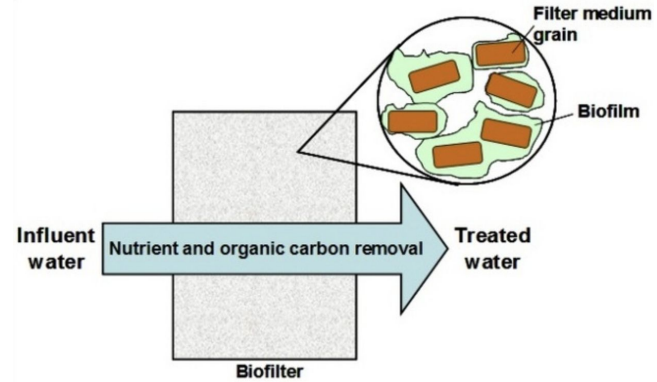
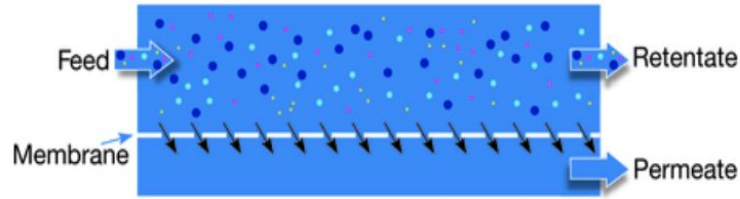
- In Scotland, 54% of sewage sludge applied to agricultural land as fertiliser.
- Seafield should prioritise research into bioremediation of MPs in sludge to mitigate long term impacts of MP pollution.

2. Current conventional treatment methods

2.1. Improving existing facility



2.2. Membrane Technology and Biofiltration

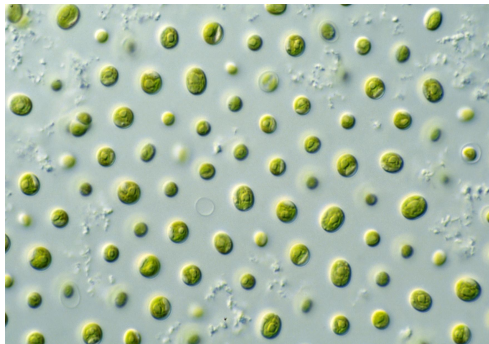


- Common type of reactors:
 - Biomembrane reactor (BMR)
 - Dynamic membrane reactor (DMR)
- High initial capital investment
- Intensive energy demand
- Membrane fouling presents high risk

2.3. Bioremediation and its challenges

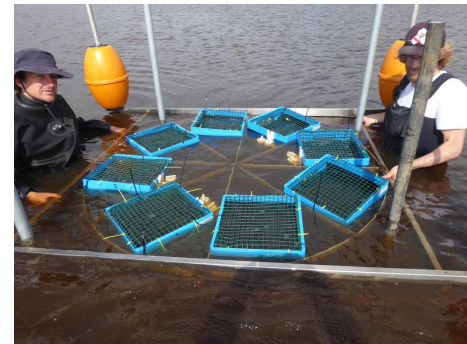
Microorganisms

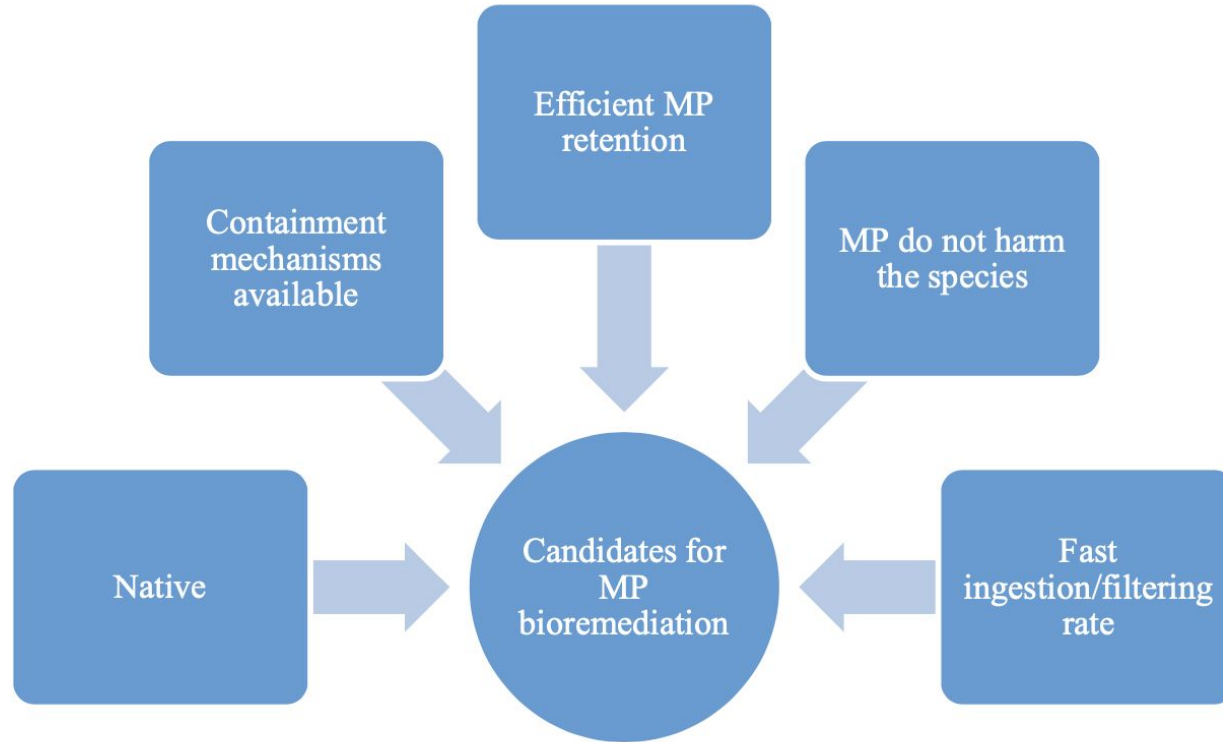
- Easy to cultivate
- Fewer ethics involved
- Can extract key enzymes
- Hard to contain



Larger Eukaryotes

- Easy to contain
- Filtration based on life-strategy
- Need to comply with welfare legislation (European Directive 2010/63)





3. Seagrass as alternatives

3.1. Key Mechanisms

2020 study by Jones et al. in Deerness Sound, Orkney, Scotland on Eelgrass (*Zostera marina*)

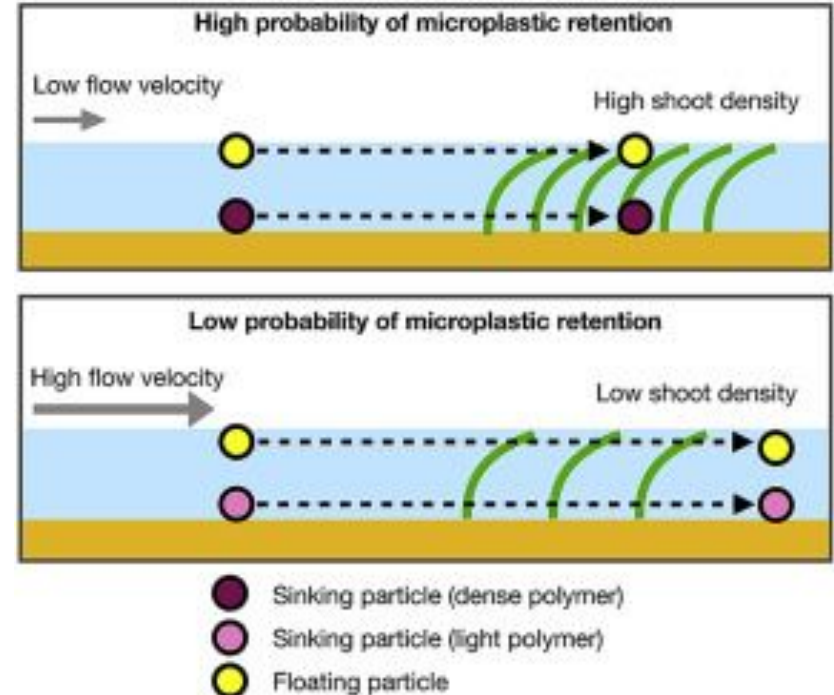
- MPs adhere to blades surface by encrustation or adhesion to biofilm.
 - 4.25 ± 0.59 MP per blade (mean $\pm$ SE, n = 60)
 - Found on all blades
 - No significant difference observed between the length of the seagrass blades analysed and the number of microplastics recorded adhering to them.
- Shows greater retention than bare sand seabed

Can be grown in sludge water (Nuzzo et al., 2020)

3.1. Key Mechanisms

Eelgrass in Gothenburg, Sweden
(Santos, et al. 2021)

- Probability of MP retention generally increased when
 - Canopy shoot density $\uparrow$
 - Specific density of the particles $\uparrow$
 - Flow velocity $\downarrow$
- Barrier created by the top part of the canopy touching the water surface stops floating MPs.



3.2. Possible implementations

Most MP are released to the environment through sludge.

- Not much options currently to remove MP from sludge and earlier effluents
- Include the use of seagrasses in the sludge production process. (at what stage?)

We assumed sea grass's capability of growing in sludge and its MP trapping/ adhering properties could be a possible solution.

Contained and monitored, periodically replaced

- To prevent accidental ingestion
- Extraction when its MP retention capacity is almost full.

3.3. Analysis

Positives

- Seagrasses are cheap and relatively easy to implement.
- Low energy demand and produce no waste or emissions.
- Natural method for removing a non-natural product, no use of chemical or energy intensive methods
- Immobile, easier to contain than most lower-order Eukaryotes solutions (microalgae)

Negatives

- Research into bioremediation using seagrasses is limited
- Risks microplastics being introduced to food chain through feeding

3.4. Future action points

Seafield will have to investigate their most abundant MP's in the sludge they produce.

Allows them to find the most effective method (Density, Flow rate and Sludge conditions) for removal of MP's.

Questions